

	Shell side	Tube side
Media	Syn Gas	CW
MOC details	Shell- SA 516 Gr 70	T.S.: SA 266 Cl 2
Test Pressure	255 Kg/Cm2	9 kg/cm2
Design pressure	170 Kg/Cm2	6 kg/cm2
Operating Pressure	134 Kg/cm2	3.1 Kg/cm2
Design Temp	140°C	60°C

Quality Assurance Plan- for Retubing of heat exchanger

Equipment: E-435 (No of tubes: 430, MOC- SA 210 Gr. C) **Size-**19.05 mm **Thickness-**2.4mm. **Tubesheet thickness-** 200 mm. It is planned to use same MOC tubes

Sr.No.	Description of activity	Type of check	Quantum of Check	Supplier	Yara/TPIA	Remarks
1	Pre inspection & testing					
	DHT	DHT	100%	Perform	Process review	Tube sheet DHT shall be carried out. The tube sheets shall be adequately covered with a ceramic blanket to ensure uniform heat soaking. Heating shall be applied using an electric coil. a) Loading temp-150°C b) Heating rate- 75°C-100°C/hr c) Soaking temp-280°C to 300°C (Max 300 °C) d) Soaking time- 8 hrs (Tube sheet thickness-200mm, Refer Note-1) e) Cooling rate-75°C-100°C/hr f) Unloading temp-150°C Note-1: The actual thickness of the tube sheet is to be verified prior to DHT, and the soaking time may be adjusted accordingly.
1.1	Simulation Test for TTS joint	Visual,DPT,Pnematic Test, pull out test, weldment sectioning tests	100%	Perform	Review	1. A minimum of 10 number of tubes shall be utilized for the simulation. 2. The material of construction (MOC) for both the tubes and the tube sheet shall be identical to the existing material. 3. In addition, pull-out tests and weldment sectioning tests shall be conducted. The Tube-to-Tube Sheet (TTS) joint shall be expanded prior to dispatching the sample to the laboratory. 4. Simulation welding shall also be carried out in accordance with the welding specification provided below.
1.2	Welder qualification test	simulation test	100%	Perform	Witness	1. Welding shall be carried out by the certified welders only. 2. The arrangement of the Welder Qualification Test (WQT) shall be within the supplier's scope. Yara (CMC) shall witness the WQT and retain the right to approve or reject the welders. 3. The WQT shall be conducted no earlier than 50 days prior to commencement of work.
1.3	Filler wire inspection	Visual inspection	100%	Perform	Review	Size,Type and packing. Make of the filler: ESAB, KOBELCO.
1.4	Tubes inspection at GR stage.	Visual and checking of Test certificates	100%	Perform	Review	1. Tubes shall be verified against the Purchase Order (PO) test certificates. 2. Inspection shall be carried out to identify any transit-related damages. 3. Inspection shall also include verification for rusting. In cases of significant rusting, chemical cleaning shall be performed as required.
2.1	Tube sheet Grinding operation	Visual	100%	Perform	-	1. The tube sheet shall remain free from any damage. 2. An emery flap disc with a grit range of 45 to 60 shall be used for the grinding operation. 3. A Dye Penetrant (DP) test shall be conducted upon completion of the grinding operation.
2.2	Checking hardness of Tube sheet	Visual	100%	Perform	Review	Hardness values should not exceed 200 BHN



3.1	Chamfering of the tube holes in the tube and cleaning of the tube holes.	Visual, Measurement	100%	Perform	-	Chamfer of 1.5 mm to 2.0 mm is to be ensured
3.2	Tube projection to be checked	Measurement	100%	Perform	-	A projection of 3 mm of the tube from the tube sheet shall be ensured.
3.3	Cleaning inspection	Visual	100%	Perform	-	Cleanliness and dryness of the tube sheet holes and the tube outer diameter (OD) shall be ensured prior to commencement of root welding.
3.4	Preheating	Preheat	100%	Perform	Process review	Preheating- Up to 150°C (Heating shall be applied using an electric coil.)
3.5	Tube to tube sheet root welding	-	100%	Perform	Process review	a) Welding Process: GTAW b) Filler Material: ER70S2, Diameter 1.6 mm, Make: ESAB / KOBELCO c) Welding Current: Maximum 90 Amp d) Upon completion of the Tube-to-Tube Sheet (TTS) welding joint, the weldment shall be immediately covered with an asbestos blanket to prevent rapid air cooling. e) Internal metal burnishing of the tubes shall be carried out up to 4 inches from the entry point.
3.6	Tube to tube sheet root welding Inspection	Visual	100%	Perform	-	Visual inspection of the root welding shall be carried out.
3.7	Tube to tube sheet final welding	-	100%	Perform	-	Same as point no: 2.9
3.8	Tube to tube sheet final welding Inspection	Visual, DPT	100%	Perform	-	Both visual inspection and Dye Penetrant Testing (DPT) of the final weld shall be performed.
3.9	Hardness Survey of the tubesheet before ISR	Hardness Measurement	100%	Perform	-	Hardness values should be recorded prior to ISR, Max value : 250 BHN.
3.10	ISR	-	100%	Perform	Process review	The tube sheets should be adequately covered with a ceramic blanket to ensure uniform heat absorption. Heating shall be applied using an electric coil. • Loading temp-150°C • Heating rate- 75°C-100°C/hr • Soaking temp-300 to 350°C • Soaking time- 4 to 6 hrs (Tubesheet thickness-200mm) • Cooling rate-75°C-100°C/hr • Unloading temp-150°C
3.11	Hardness Survey of the tubesheet after ISR	Hardness Measurement	100%	Perform	-	Hardness values should be recorded after ISR, Range : 200 to 220 BHN.
3.12	Pneumatic Test @ 2.0 Kg/cm ² before expansion (or as per re-tubing procedure given in the equipment drg)	Soap solution test	100%	Perform	Witness	No air leakage to be allowed
3.13	Expansion of TTS joints	Visual	100%	Perform		1. Tubes shall be expanded within the tube sheet holes using a swage-type expansion tool, rather than a tapered rotating tool 2. Inspection to be done using bore gauge
3.14	Pneumatic Test @ 2.0 Kg/cm ² after expansion (or as per re-tubing procedure given in the equipment drg)	Soap solution test	100%	Perform	Witness	No air leakage to be allowed
3.15	Helium Leak test	Helium Leak test	100%	Perform	Witness	Perform Helium Leak test after tube expansion. Max acceptable leak rate 1X10 ⁻⁵ Std CC/sec (Leak rate 1X10 ⁻⁷ Std CC/sec shall be considered as No leakage condition). The helium leak test can be performed at @ 2Kg/cm ² g (same as pneumatic test pressure) with min. 10% concentration (by volume) of Helium gas.
3.16	Hydro testing	Visual	100%	Perform	Witness	1) Hydrotesting shall be conducted at a pressure of 212.5 kg/cm ² on the shell side (1.25 times the design pressure is considered instead of 1.5 times, due to the prolonged service life of the equipment). 2) Holding time: 30 minutes. 3) Chloride concentration in the water shall be maintained below 20 ppm.

